Synthetic fixture report

Free fall

Vertical position against time; gravity comes from the curvature of the fitted polynomial.

- Motion is vertical with negligible drag.
- Time and position share one origin.

Results

Fitted g is 9.80246 ± 0.0777081 m/sˆ2 (simulated) from 8 (simulated) samples; residual RMS is 0.00232037 m (simulated).

This is a simulated fixture, not a physical measurement.

Table 1: Reported quantities (values copied from fitter output)

Quantity	Value	Unit	Uncertainty (same unit)	Source label
g	9.80246	m/sˆ2	0.0777081	simulated
n	8	1		simulated
rmse	0.00232037	m		simulated

Table 2: Fit residuals per sample

Sample	Fitted	Residual	Unit	Source label
0	0.998266	0.001734	m	simulated
1	1.08898	2.02835e-05	m	simulated
2	1.2801	-0.0041004	m	simulated
3	1.56113	-0.000127998	m	simulated
4	1.94494	6.48285e-05	m	simulated

Table 2: Fit residuals per sample

Sample	Fitted	Residual	Unit	Source label
5	2.42146	0.00453663	m	simulated
6	3.00662	-0.000623225	m	simulated
7	3.6845	-0.00150411	m	simulated

fitter_digest sha256:024bc80b4466b105ff8d375b86a5a06180febfad1a89e782c8c3f5d480e6037c; labels: simulated